

Notes on

Group Theory

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This text consists of notes on the course Group Theory, lectured by Professor Sujit Kumar Sardar at Jadavpur University in the 5th semester 2025.

Some changes and some additions have been made by me. To distinguish them from the lecture's actual contents, they are labelled with an asterisk. So any *Lemma** or *Remark** or *Proof** that the reader might come across are wholly my responsibility.

Please report errors, typos etc. to this email: dassayan0013@gmail.com.

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List of Lectures

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Course organisation. — The lectures will take place on Mondays from from 12:00 to 14:00 and on Thursdays from 14:30 to 16:30, in the PG-I classroom on the ground floor of Ramanujan Bhavan.

The prerequisite to this lecture is a course on groups, rings and fields upto the first isomorphism theorem, akin to the one covered in the second semester course [Algebra](#). For this course, the default recommendation is Dummit and Foote ([\[DF04\]](#)). Professor Sardar will also occasionally refer to the text by Hungerford ([\[Hum12\]](#)). I personally really like the text by Jacobson ([\[Jac09\]](#)).

The syllabus of this course is available [here](#).

CHAPTER 1.

1

Isomorphism theorems and automorphisms

1.1. Introduction

LECTURE 1
4th Aug, 2025

1.1.1. Isomorphism of groups. — *How many examples of groups can you think of ?*
Answers from the class included

- (under the operation of addition) $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
- (under the operation of multiplication) $\mathbb{Q}^*, \mathbb{R}^*, \mathbb{C}^*$ (where $X^* = X \setminus \{0\}$)
- the matrix groups $\mathrm{GL}_n(\mathbb{R}), \mathrm{SL}_n(\mathbb{R}), \mathrm{O}(n), \mathrm{SO}(n)$ under multiplication
- the integers modulo n , $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ under addition (modulo n of course)
- given any two groups we can also construct a group by taking their Cartesian product, such as $\mathbb{Z}_n \times \mathbb{Z}_n$, called the *direct product*

Is it possible to construct a group out of the following elements: $\{a, b, \$, \#\}$?

These are *abstract* symbols, which a priori don't mean anything, as opposed to something *concrete* like, say, the integers modulo 4, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$. We can still define a group as long as we define a *binary operation* $G \times G \rightarrow G$, $(g_1, g_2) \mapsto g_1 g_2$ on the given set satisfying the following 3 properties:

- (1) associativity: $(g_1 g_2 g_3) = g_1 (g_2 g_3) \quad \forall g_1, g_2, g_3$
- (2) existence of identity: $\exists e : e g_1 = g_1 e = g_1 \quad \forall g_1$
- (3) existence of inverse: $\forall g_1 \exists g_1^{-1} : g_1^{-1} g_1 = g_1 g_1^{-1} = e$

and we can define such an operation by identifying our 4 abstract symbols with the elements of some group of order 4. This identification is made precise using the language of *isomorphism*.

1.1.2. Definition*. — Suppose we have two algebraic structures $(A, \spadesuit)$ and $(B, \diamond)$ equipped with the binary operations $\spadesuit$ and $\diamond$ respectively.

A **homomorphism** is a *structure-preserving* mapping $f : A \rightarrow B$ between A and B i.e.

$$f(a \spadesuit b) = f(a) \diamond f(b) \quad \forall a, b \in A.$$

An **isomorphism** is a bijective homomorphism. If A is isomorphic to B , denoted $A \cong B$, it means that there's no distinction between A and B except the names of the elements.

1.1. INTRODUCTION

1.1.3. Groups of order 4. — How many groups of order 4 are there ?

Immediately one thinks of $(\mathbb{Z}_4, +)$, the Klein 4-group $(K_4, \cdot)$,¹ $(\mathbb{Z}_2 \times \mathbb{Z}_2, +)$, the group of 4th roots of unity $U_4 = \{a \in \mathbb{C} : a^4 = 1\} = \{\pm 1, \pm i\}$ under multiplication and so on. Observe that $\mathbb{Z}_4 \cong U_4$ while $K_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Indeed, upto isomorphism we have exactly 2 groups of order 4. When we say “# {groups of order n }”, we mean upto isomorphism.

To see why $\mathbb{Z}_4 \cong U_4$, we can construct the Cayley tables for both and compare them

	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

The highlighted rows are the generators $\{1, 3\}$ for $\mathbb{Z}_4$ and $\{\pm i\}$ for U_4 . Now recall that a cyclic group of order n has exactly $\phi(n)$ generators, and $\phi(4) = 2$. This means there are no other generators for each of these two groups. Then $U_4 \cong \mathbb{Z}_4$ by an explicit isomorphism

$$i \mapsto 1, -i \mapsto 3, -1 \mapsto 2, 1 \mapsto 0.$$

Similarly, K_4 abstractly is given by the relations

$$a^2 = b^2 = c^2 = e, ab = ba = c, cb = bc = a, ac = ca = b$$

or by the following Cayley table, showing that it is non-cyclic.

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Consider the concrete group $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. Then $K_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ is given explicitly by

$$e \mapsto (0, 0), a \mapsto (0, 1), b \mapsto (1, 0), c \mapsto (1, 1).$$

Observe how no element repeats in the same row or column for the Cayley tables we've seen so far. This holds in general. *Why ?*

Suppose $b \cdot a = d = b \cdot c$ for two different rows (or columns). Then multiplying by the inverse of b on both sides (inverse of b exists by the group axioms) we get $a = c$. But $a \neq c$ as these are different rows (or columns), hence an absurdity.

This essentially turns the task of filling the Cayley table of a group into a sudoku puzzle. We are now ready to give an elementary proof that there are only 2 groups of order 4. The idea is to begin with an abstract set of 4 elements $\{1, a, b, c\}$, and then to try and construct all possible composition tables for this set such that the composition satisfies the 3 properties required for a group. We will find that there are only 2 tables upto isomorphism.

Proof. Consider the set $\{1, a, b, c\}$ taking 1 to be the identity. Then the possible Cayley tables are reduced to the following four possibilities by solving the sudoku puzzle: the 1st row and column remain invariant and no element should repeat in the same row or column

¹also known as the Vierergruppe V from the German *vier*, “four”

1.1. INTRODUCTION

	1	<i>a</i>	<i>b</i>	<i>c</i>		1	<i>a</i>	<i>b</i>	<i>c</i>		1	<i>a</i>	<i>b</i>	<i>c</i>		1	<i>a</i>	<i>b</i>	<i>c</i>
1	1	<i>a</i>	<i>b</i>	<i>c</i>	1	1	<i>a</i>	<i>b</i>	<i>c</i>	1	1	<i>a</i>	<i>b</i>	<i>c</i>	1	1	<i>a</i>	<i>b</i>	<i>c</i>
<i>a</i>	<i>a</i>	1	<i>c</i>	<i>b</i>	<i>a</i>	<i>a</i>	<i>b</i>	<i>c</i>	1	<i>a</i>	<i>a</i>	<i>c</i>	1	<i>b</i>	<i>a</i>	<i>a</i>	1	<i>c</i>	<i>b</i>
<i>b</i>	<i>b</i>	<i>c</i>	1	<i>a</i>	<i>b</i>	<i>b</i>	<i>c</i>	1	<i>a</i>	<i>b</i>	<i>b</i>	1	<i>c</i>	<i>a</i>	<i>b</i>	<i>b</i>	<i>c</i>	<i>a</i>	1
<i>c</i>	<i>c</i>	<i>b</i>	<i>a</i>	1	<i>c</i>	<i>c</i>	1	<i>a</i>	<i>b</i>	<i>c</i>	<i>c</i>	<i>b</i>	<i>a</i>	1	<i>c</i>	<i>c</i>	<i>b</i>	1	<i>a</i>

where the first table (in red) is just the group K_4 and the rest three are isomorphic to $\mathbb{Z}_4$. Clearly $\mathbb{Z}_4 \not\cong K_4$ as in K_4 every element is its own inverse but in $\mathbb{Z}_4$ we have 0 and 2 are their own inverses but $1 + 1 = 2$ and $3 + 3 = 2 \pmod{4}$ are not their own inverses. Hence, upto isomorphism, there are only 2 groups of order 4. $\square$

1.1.4. Question. — Find a group of 4 invertible matrices isomorphic to the Klein 4-group.

Solution. The following group of invertible matrices of order 4,

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$$

with identity $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is isomorphic to the Klein-4 group as

$$e \mapsto \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad a \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b \mapsto \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad c \mapsto \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

A more natural solution is given by *permutation matrices* that we'll see soon enough. $\square$

There is nothing special about 4. We can ask the more general question

1.1.5. Question (Classification problem). — How many groups (upto isomorphism) of order n are there ?

A closely related question is the problem of representing a finite group using matrices

1.1.6. Question (Representation problem). — Is every group of order n isomorphic to a group of invertible matrices ?

1.1.7. Question. — Suppose G is a group of order n . Find an example of a group G' of order n whose elements are permutations of a suitable set such that $G \cong G'$. In particular, what about K_4 and $\mathbb{Z}_4$?

Permutations are bijections of a nonempty set onto itself. The set need not be finite.

1.1.8. Definition/Proposition (Symmetric group). — Let $X \neq \emptyset$. We denote by S_X or $\text{Sym}(X)$ the set of all permutations on X . Then $\text{Sym}(X)$ is a group under function composition. The group $\text{Sym}(X)$ is called the **symmetric group** and its subgroups are called **permutation groups**.

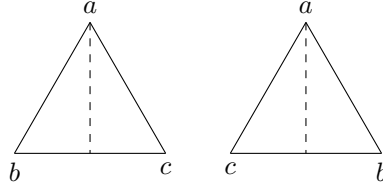
The name “symmetric group” comes from the fact that historically these were studied as the group of symmetries of the roots of polynomials. In fact, these were the first types of groups that were studied (such as by Abel, Lagrange and Galois). “Symmetry” evokes the geometric notion of the same name. It is natural to ask how many symmetries does a shape have, but how can we make this precise ?

We consider *rigid motions* or *transformations* that preserve distances between any two points, don't cause any deformation. Then a symmetry is a rigid transformation under which the shape remains invariant.

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1.1.9. Definition (Symmetry). — A **symmetry** of a shape is a permutation of its vertices.

1.1.10. Example. — Mirror reflection is a symmetry of a triangle



which is represented by the following permutation of vertices

$$\begin{pmatrix} a & b & c \\ a & c & b \end{pmatrix}.$$

1.1.11. Theorem (Cayley's theorem). — *Every group is isomorphic to a subgroup of the symmetric group.*

Proof. Let G be a group and for any $a \in G$ define the map $a_L : x \mapsto ax$ of G into G called the *left translation/multiplication by a* . Then $a_L : G \rightarrow G$ is a bijection, as there exists an inverse map $(a_L)^{-1}(x) = (a^{-1})_L(x) = a^{-1}x$ such that

$$(a^{-1}a)_L(x) = (a_L^{-1}a_L)(x) = \text{id}(x) = x.$$

Hence, $a_L \in \text{Sym}(G)$. Construct $K = \{a_L : a \in G\}$. Then K is a subgroup of $\text{Sym}(G)$. Now consider the map $f : G \rightarrow K$ such that $f(a) = a_L$. Then

$$f(ab) = (ab)_L(x) = (ab)x = a(bx) = a_L(b_L(x)) = (a_L b_L)(x) = f(a)f(b).$$

Thus f is a homomorphism. Further,

$$f(a) = f(b) \implies a_L(x) = b_L(x) \implies ax = bx \implies a = b$$

so f is injective. Finally, $\forall a_L \in K \exists a \in G : f(a) = a_L$, so it is surjective, and so bijective. Thus f is an isomorphism and $G \cong K$ which is a subgroup of $\text{Sym}(G)$. $\square$

Subgroups of $\text{Sym}(G)$ and left translation by a defined in the proof of Theorem 1.1.11 provide us with a boundless source of examples and counterexamples. For instance,

Solution to 1.1.7. By Cayley's theorem 1.1.11, $G \cong G'$ where $G' = \{a_L : a \in G\}$ is a subgroup of $\text{Sym}(G)$ of order $|G| = n$. For K_4 , the required permutation group then is

$$\mathcal{G}_1 = \{e_L, a_L, b_L, c_L\} = \{e, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$$

while for $\mathbb{Z}_4$ we have the permutation group

$$\mathcal{G}_2 = \{0_L, 1_L, 2_L, 3_L\} = \{e, (1\ 3)(2\ 4), (1\ 4)(2\ 3), (1\ 2\ 3\ 4)\}.$$

$\square$

Alternate solution to 1.1.4. The group of permutation matrices

$$S = \left\{ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \right\} \quad (1.1.1)$$

is isomorphic to K_4 . $\square$

The problem 1.1.6 and the matrix group we found in (1.1.1) lead us naturally into *representation theory* - which is concerned with getting a complete idea of an abstract group by considering a matrix group “representation” isomorphic to it, such as the group (1.1.1) in the case of K_4 .

Solution to the representation problem 1.1.6. In general, by Cayley’s theorem any finite group G is isomorphic to some permutation group G' corresponding to which we have a group of permutation matrices yielding the required representation of G in terms of invertible matrices. $\square$

LECTURE 2
13th Aug, 2025

1.1.12. Cosets. — For an abstract group G we will always use the multiplicative notation for the operation unless otherwise stated. We will denote a subgroup H of G by $H \leq G$. For a normal subgroup N of G we write $N \trianglelefteq G$.

A **left coset** of H is determined by an element $a \in G$, denoted $aH = \{ah : h \in H\}$. Similarly a **right coset** is $Ha = \{ha : h \in H\}$.

1.1.13. Question. — (1) For which $a \in G$, $aH = H$?

(2) When is $aH = bH$? ($a, b \in G$)

(3) For which $b \in G$, $bH = aH$? ($a \in G$ is given)

(4) For $a, b \in G$, is the product of aH and bH meaningful ?

Is the product a left coset of H ? That is, does there exist $c \in G$ such that $(aH)(bH) = cH$?

1.1.14. Warning*. — It is not true that $(aH)(bH) = abH$ if we only know $H \leq G$. We can draw that conclusion if H is a *normal subgroup*, but not if H is just any subgroup.

Given a group G , $H \leq G$ let us define a relation L_H on G by $aL_H b \iff a^{-1}b \in H$. Then

(1) L_H is an equivalence relation on G .

(2) Let $a \in G$. The equivalence class of a is aH .

(3) For any $b \in aH$, $aH = bH$. For no element $b \in G \setminus aH$, $aH = bH$.

LECTURE 3
14th Aug, 2025

1.1.15. First Isomorphism Theorem. — $\tilde{f} : G_1 / \ker f \rightarrow \text{im } \tilde{f} = \text{im } f$.

$$\boxed{G_1 / \ker f \cong \text{im } f}.$$

$L_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ $x \mapsto Ax$ is a group homomorphism and a linear transformation. We have $\ker L_A = \{x \in \mathbb{F}^n : Ax = 0\}$ = the set of all solutions of the system (m equations and n constants) of homogeneous linear equations with coefficient matrix A .

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$L_A^{-1}(b)$ = the set of all solutions to $Ax = b$, i.e. $L_A^{-1}(b) = x_0 + \ker L_A$ where $Ax_0 = b$ for some $x_0 \in \mathbb{F}^n$. (special case: General solution = CF + PI in ODEs)

$Ax = b$, $Ax = b_1$ consistent $\implies$ set of solutions are of the same cardinality.

LECTURE 4
18th Aug, 2025

1.1.16. Inner automorphism. — Let $g \in G$. Define $i_G : G \rightarrow G$ by $i_G(x) = gxg^{-1} \forall x \in G$. Then clearly

$$\begin{aligned} i_G(xy) &= xyg^{-1} = gx(g^{-1}g)yg^{-1} \\ &= (gxg^{-1})(gyg^{-1}) \\ &= i_G(x)i_G(y). \end{aligned}$$

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